

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

M.Tech. Computer Engineering Sem - I University Examination, 2009-10

CE 704 Soft Computing

Date: 06.01.11, Thursday Time: 10:30 a.m To 01:30 p.m Maximum Marks: 70

- Instructions:**
- (1) All questions are compulsory.
 - (2) Indicate clearly, the option you attempt along with its respective question number.
 - (3) Figures to right indicate marks
 - (4) Rough work is done in the last page of main supplementary, don't write anything on the question paper.

SECTION-I

Q-1 Answer the following questions.

1. What are the distinctive characteristics of multi layer perceptrons? Enlist any five application areas where ANN will be preferably applicable. 3
2. What is bias? Compare weight and bias. What is the use of threshold function? 3
3. The input to a Single neuron is 5.0, its weight is 3.5 and bias is -2 then calculate: net input to transfer function and the neuron output. 3
4. What is McCulloch and Pitts model for an artificial neuron? 2

Q-2

- [A] Give the structure of : 4
1. Single layer feed forward NN
 2. Multilayer feed forward NN
 3. Recurrent NN
- [B] List the features of fuzzy NN. Justify the use of fuzzy logic in soft computing. What are the criticisms for Fuzzy Logic? Explain the term fuzzy distance. 4
- [C] How do we decide how many hidden layers and number of neurons in each hidden layer should be there? "Increasing the number of hidden layer neurons gives a better function approximation". Is this statement correct? Defend your answer. 4

Q-3

- [A] Where and how the knowledge is stored in NN? Explain with the help of a suitable example. How NNs are used to classify patterns for a given problem. 4
- [B] What are the characteristics of the problem that are to be analyzed when choosing an application method to solve the problem? Explain. State the difference(s) between pattern recognition and image understanding. 4
- [C] What is meant by activation function in an artificial neuron model? Describe the various activation functions that are employed and compare their merits and demerits. 4

SECTION-II

Q-4

1. Let Y and R be two fuzzy sets of young and rich people. What is the member grade of a person being young and rich if the member grade of a person being young is 0.8 & being rich is 0.7? 3
2. Explain how you can change the weights according to Hebbian learning rule. 4
3. Differentiate between competitive learning and Self-organizing feature maps. 4

Q-5

- [A] An automatic washing machine controls the washing cycle speed based on the quantity and quality of clothes. If the clothes are soft and quantity is less then washing cycle is low and vice versa. Derive fuzzy logic control system that controls washing cycle based on clothes quantity and quality. Define fuzzy variables and fuzzy rules. 4
- [B] Why there is a need for hybridization of various soft computing techniques? How GA play important role with integration of other technology? Explain in details. 4
- [C] Describe how knapsack problem can be solved using Genetic algorithm. Describe your solution using suitable example. 4

OR

- [A] Explain how GA is useful for solving nonlinear optimization problem? What are the stopping criteria while executing GA? 4
- [B] How selection of candidates for mating pool will affect convergence of algorithm? Why mutation is required? 4
- [C] What is Hopfield network? Why do we say that the network performs an associative memory? Give features of Hopfield network known as theory of memory. Can it be possible to use Hopfield Network for error correction? 4

Q-6

- [A] How is Boltzman learning rule applied to Energy minimization? 4
- [B] How does the ART network overcome the stability-plasticity dilemma? 4
- [C] Define Encoding System in GA. Write the Encoding representation of transportation Problem. 4

OR

- [C] The following table consists of training data from an employee database. The data have been generalized. For a given row entry, count represents the number of data tuples having the values for department, status, age and salary given in the row. 4

Department	Status	Age	Salary	Count
Sales	Senior	31..35	46K..50K	30
Sales	Junior	26..30	26K..30K	40
Sales	Junior	31..35	31K..35K	40
Systems	Junior	21..25	46K..50K	20
Systems	Senior	31..35	66K..70K	5
Systems	Junior	26..30	46K..50K	3
Systems	Senior	41..45	66K..70K	3
Marketing	Senior	36..40	46K..50K	10
Marketing	Junior	31..35	41K..45K	4
Secretary	Senior	46..50	36K..40K	4
Secretary	Junior	26..30	26K..30K	6

Let the salary be the class label of the attribute.

1. Design a multi-layer feed-forward network for the above data. Labels the nodes in the input and output layers.
2. On which attribute can fuzzy set approach be used? Mention the fuzzy values for these attributes and write the methodology to be used.
3. Which learning algorithm would you prefer? What kind of Activation function should be used? What are the default parameters you consider?